

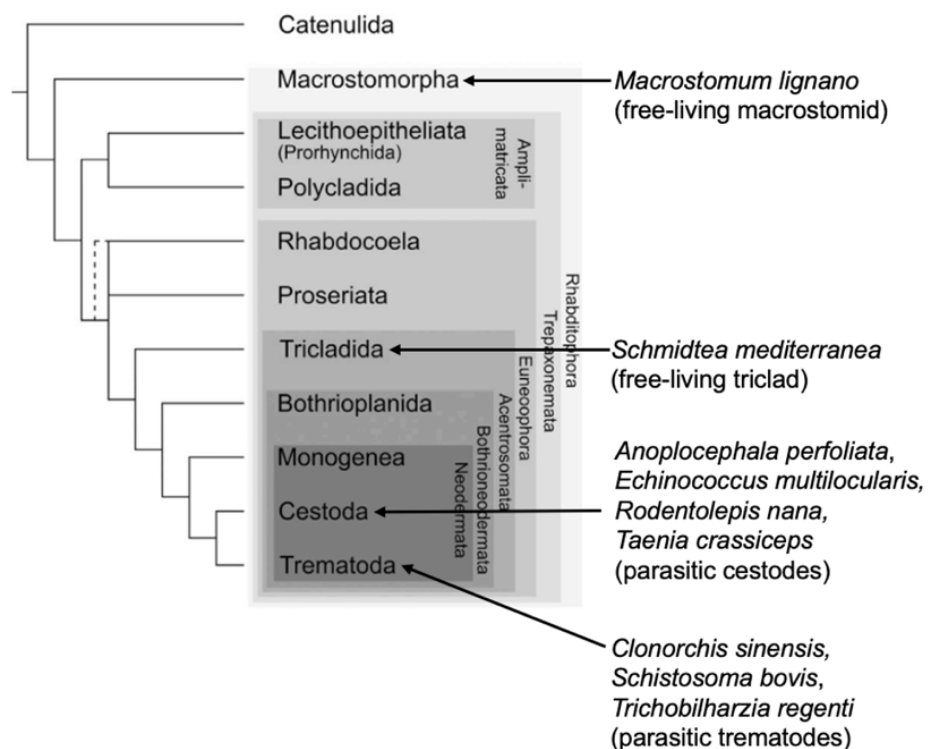
CANDIDATE CORRECTIONS TO MANUSCRIPT

- [Page 4]:
 - Figure 1: Phylogeny of metazoan species from which **HIF pathway gene** anchor sequences etc.

Protein sequences from 25 species across 13 animal phyla were used to construct **p-HMMs for HIF- α , EGLN and VHL ortholog mining** across the proteomes of 9 platyhelminth species.

Priapulida clades with Ecdysozoa, not Lophotrochozoa.

- [Page 5]:
 - Figure 2: **AMENDED TO**



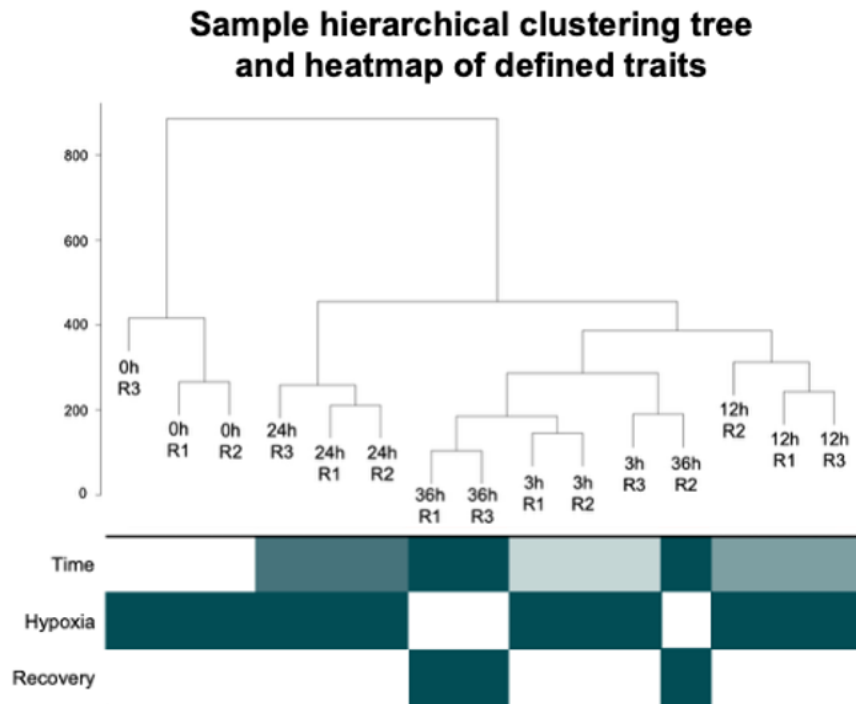
- [Page 10]: **FINAL PARAGRAPH AMENDED TO**

ATAC-seq consensus peakset construction

ATAC-seq consensus peakset construction was performed using DiffBind (Stark & Brown, 2011). A consensus peakset was subsequently generated using dba.count with minOverlap = 2, retaining peaks present in at least two libraries. Peak summits were re-centred using a 250 bp summit extension.

- [Page 14]:
 - **For HIF- α ortholog mining**, alignment using MAFFT iterative local-pair refinement was selected to maximise positional homology etc.
- [Page 16]:
 - A single candidate VHL ortholog in the **trematode** *Clonorchis sinensis*.
 - Subsequent maximum-likelihood phylogenetic inference placed the *Clonorchis* sequence **in a phylogenetic tree of high-confidence metazoan VHL sequences** as an isolated branch outside other metazoan VHL clades etc.

- [Page 21]:
 - Figure 11A: **AMENDED**



- Figure 11C: Heatmaps **beside** the module assignments etc.
- Figure 11D: Temporal trajectories of module expression showing mean FPKM **value** of genes etc.
- [Page 31]: These indirect sources of evidence are critical in a non-model metazoan with no transgenic technologies available, precluding the use of **CRE-reporter** constructs for directly testing the function and mapping the target genes of CREs.
- [Page 32]: Figure 18: Base-pair resolution view of hypoxia-specific Klf10/11 footprint in **h1SMcG0021003**.